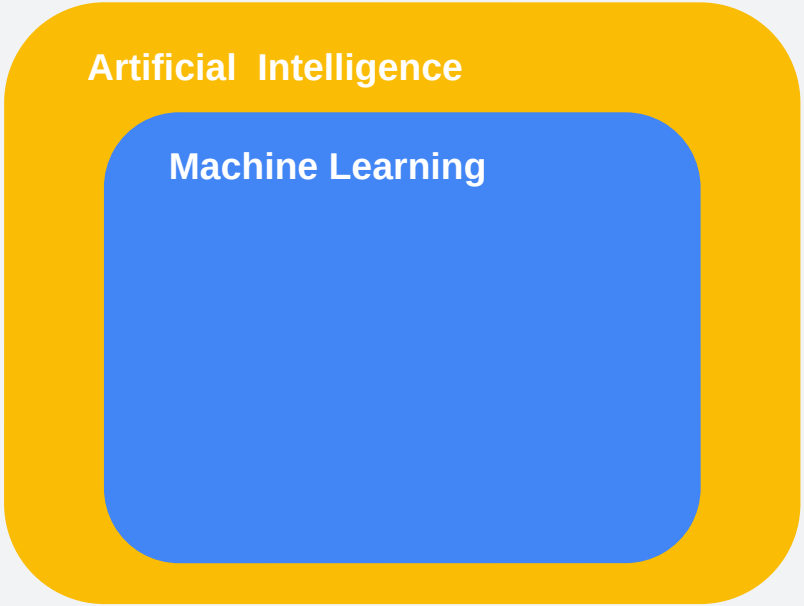
What is (tiny) Machine Learning?

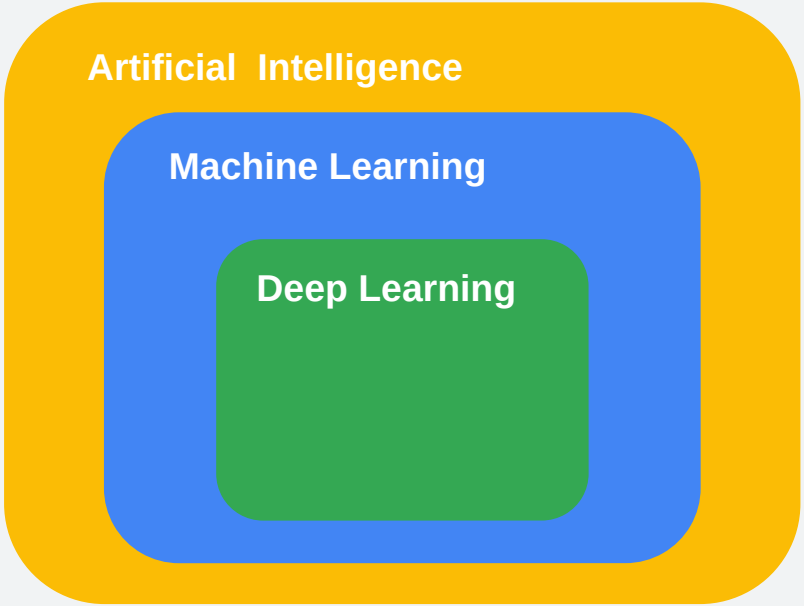
What is Machine Learning?

1. **Machine Learning** is a subfield of **Artificial Intelligence** focused on developing algorithms that learn to **solve problems by analyzing data for patterns**



What is (Deep) Machine Learning?

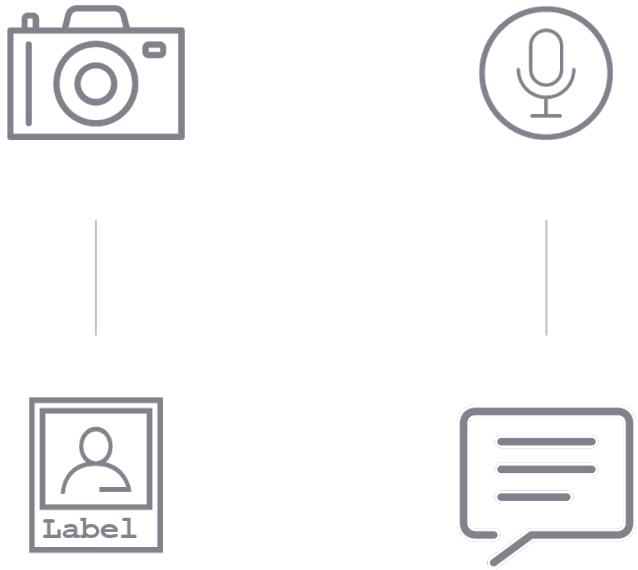
1. Machine Learning is a subfield of Artificial Intelligence focused on developing algorithms that learn to solve problems by analyzing data for patterns
2. **Deep Learning** is a type of Machine Learning that leverages **Neural Networks** and **Big Data**



Applications of Machine Learning



Applications of Machine Learning



Applications of Machine Learning

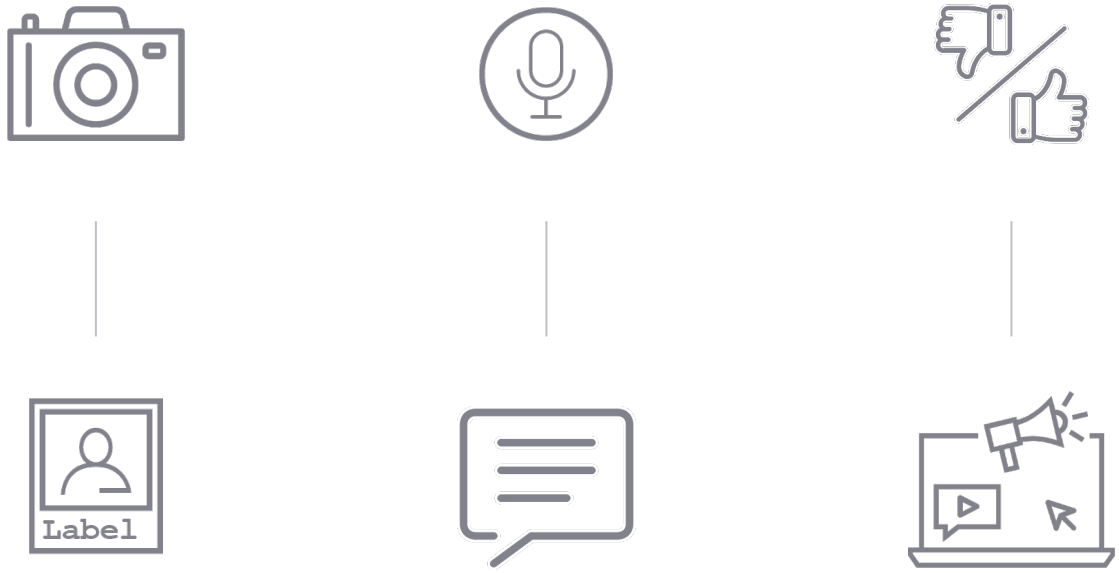
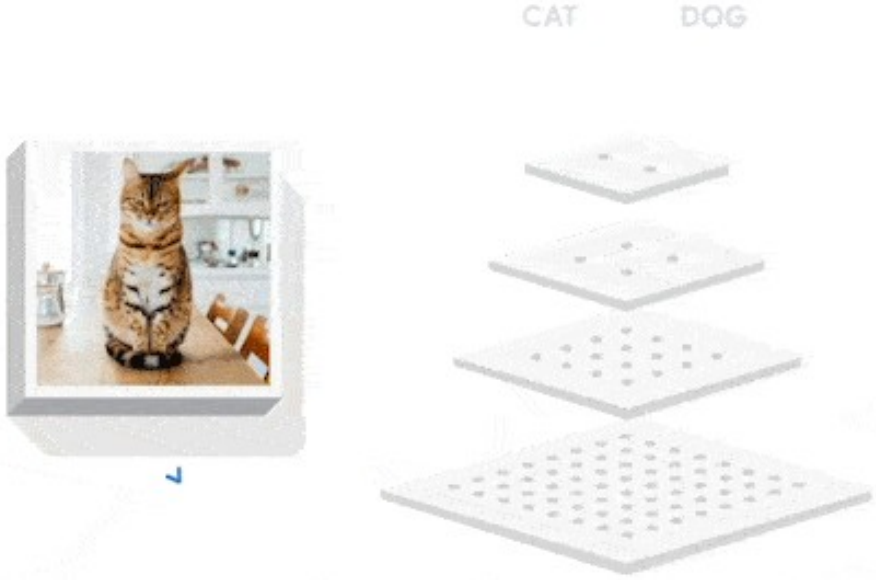
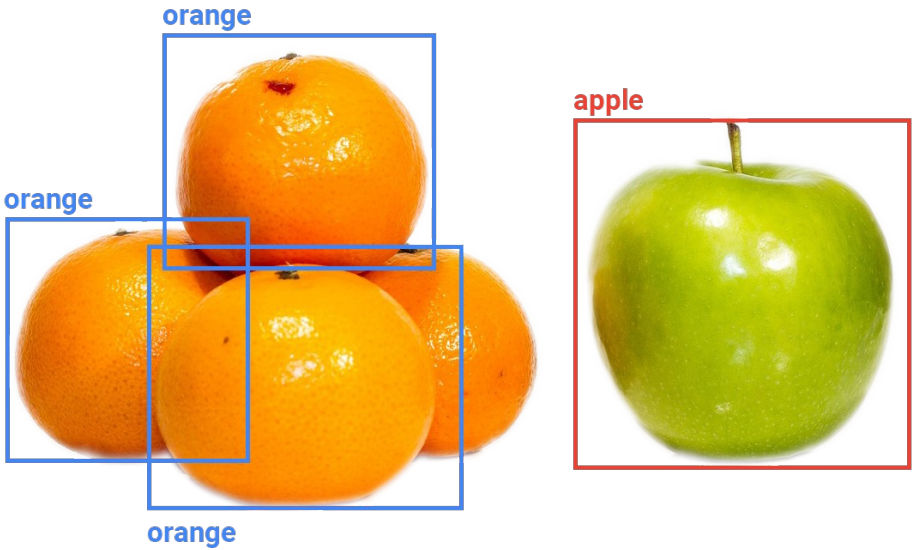


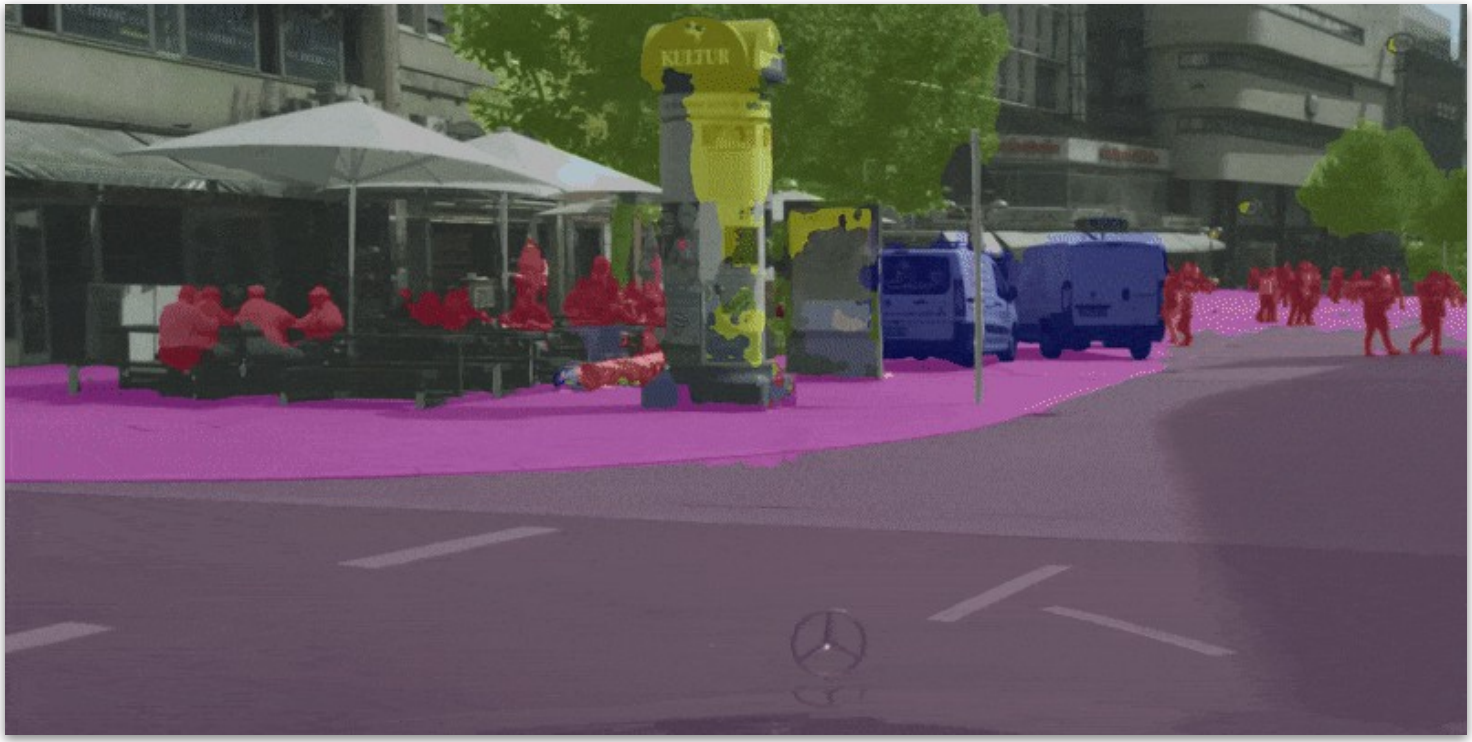
Image Classification



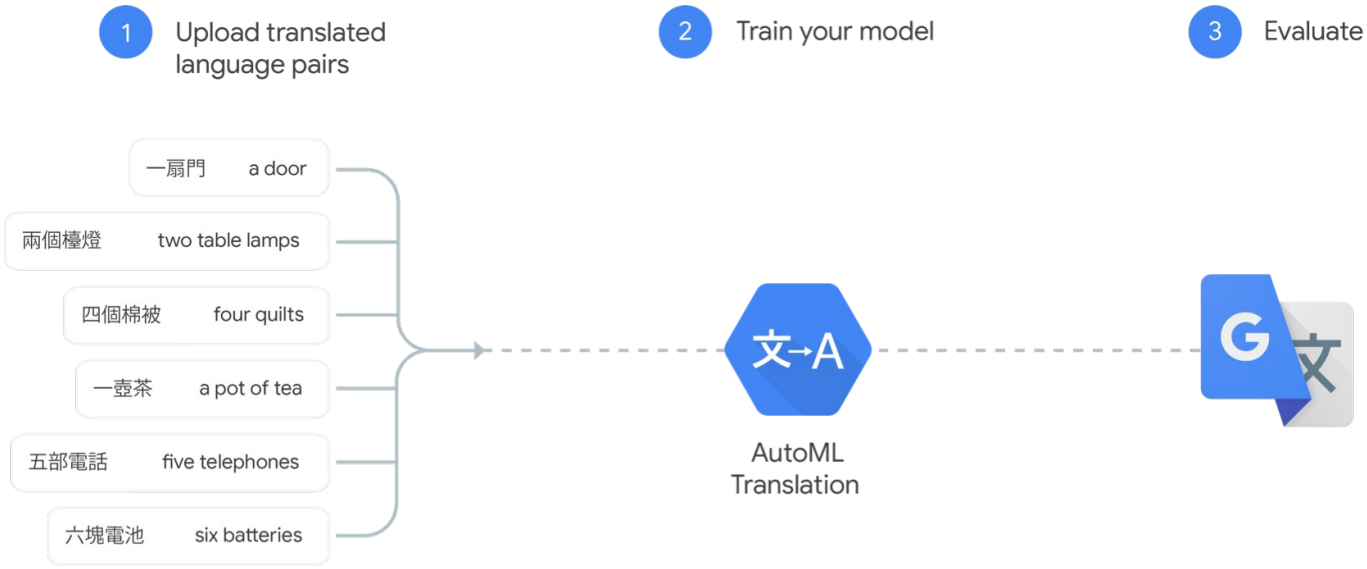
Object Detection



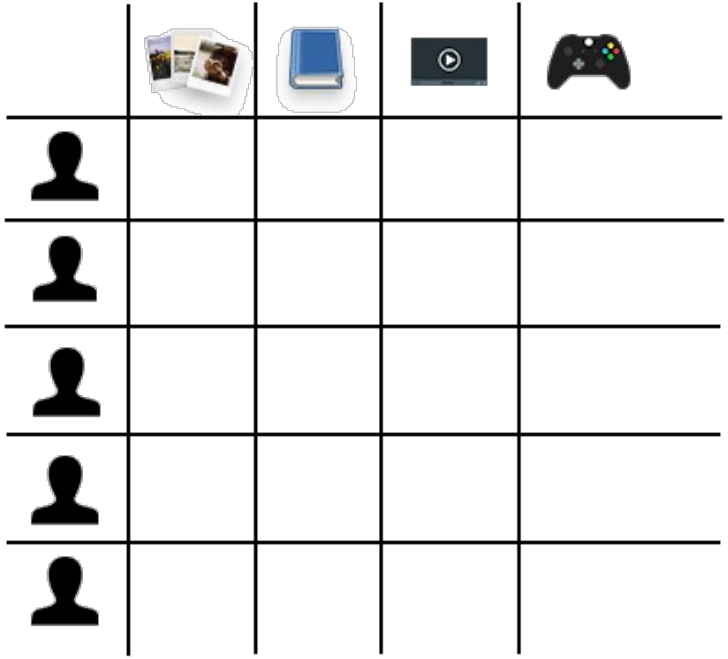
Segmentation



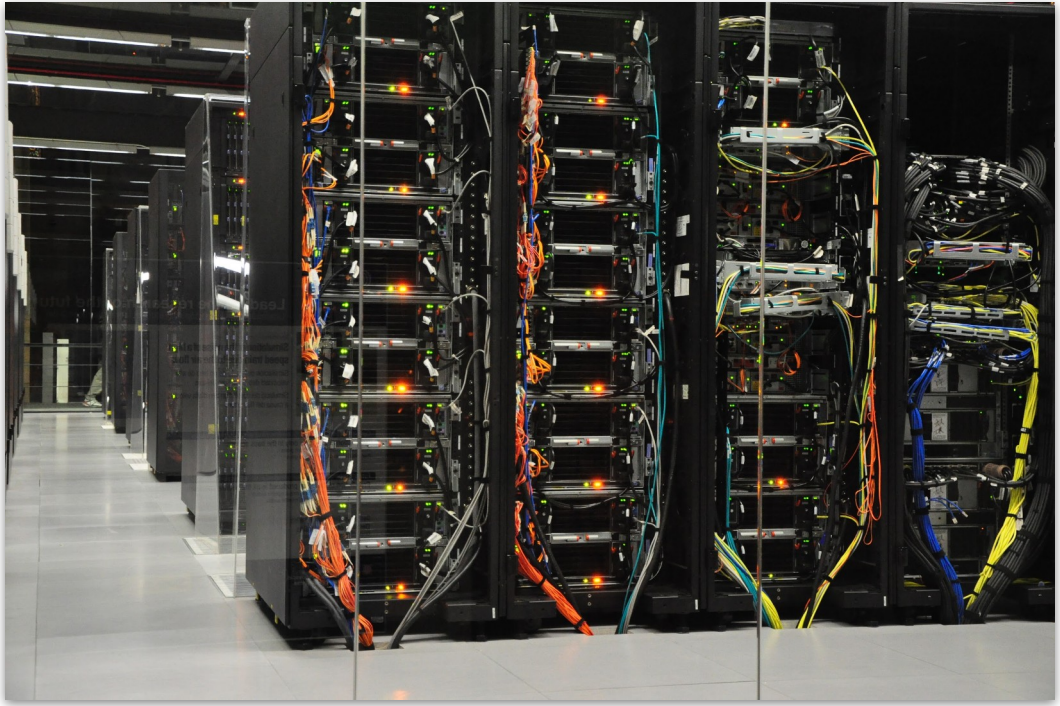
Machine Translation



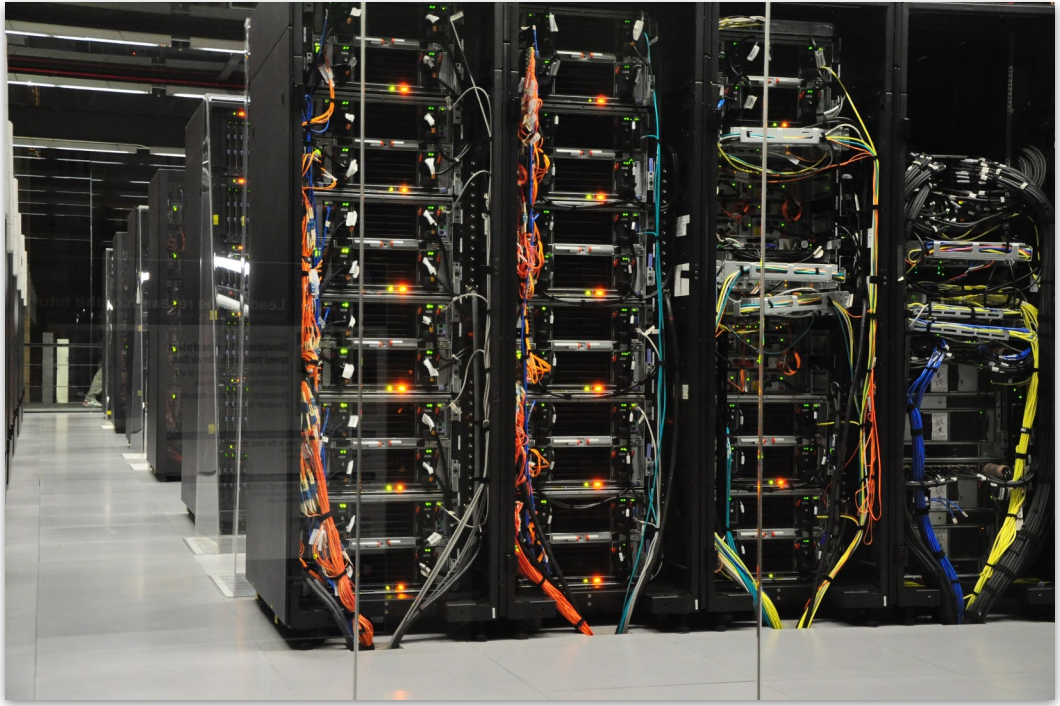
Recommendations



Datacenter

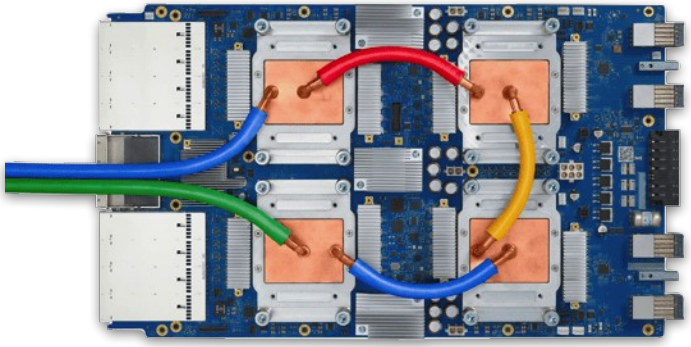


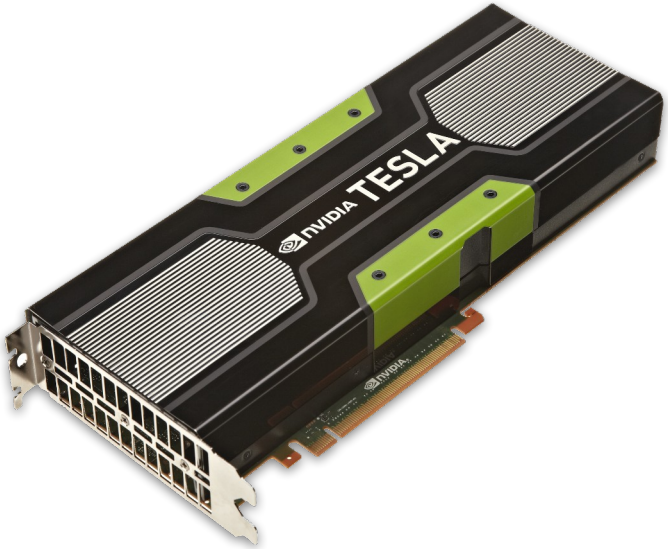
Datacenter





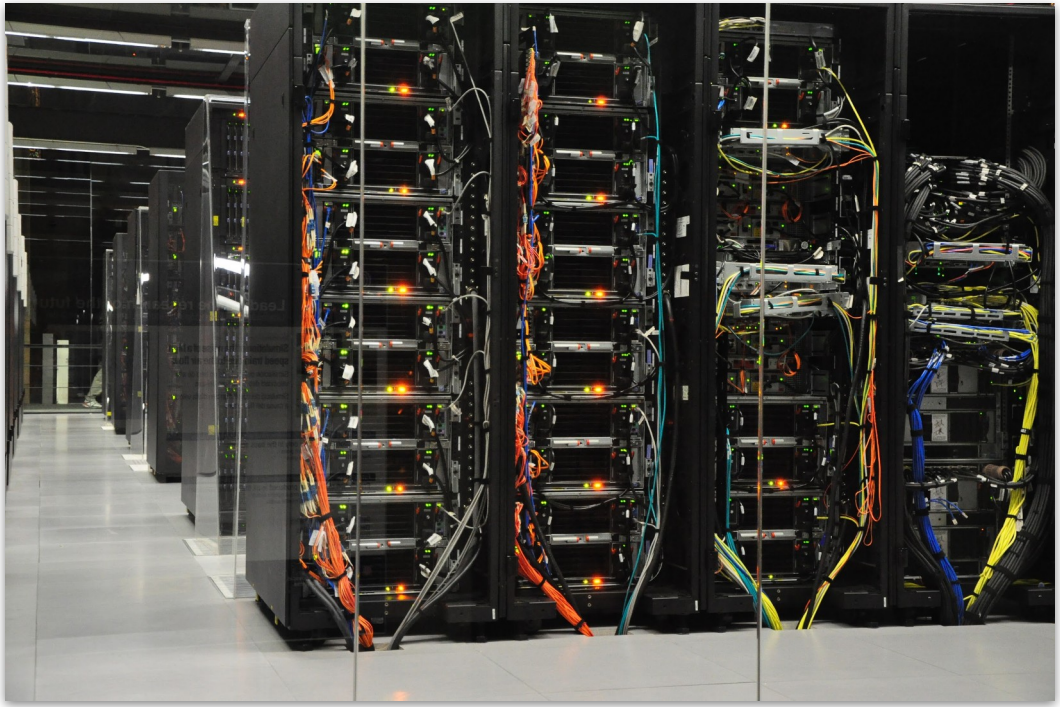
TPUs/GPUs

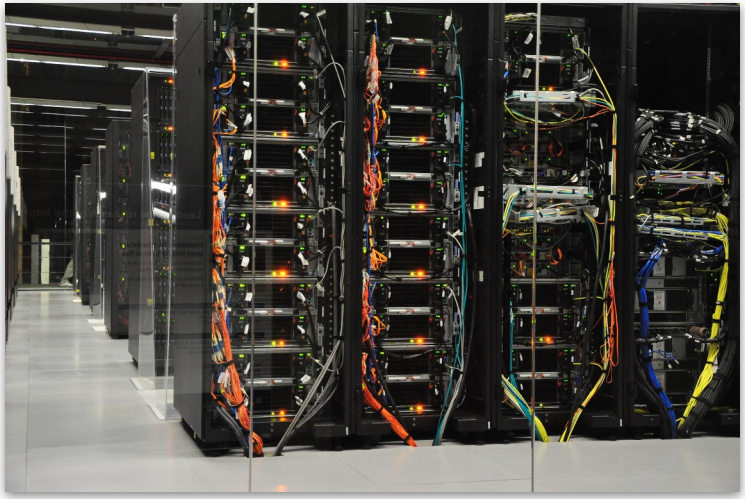






Bigger Is Not Always Better.



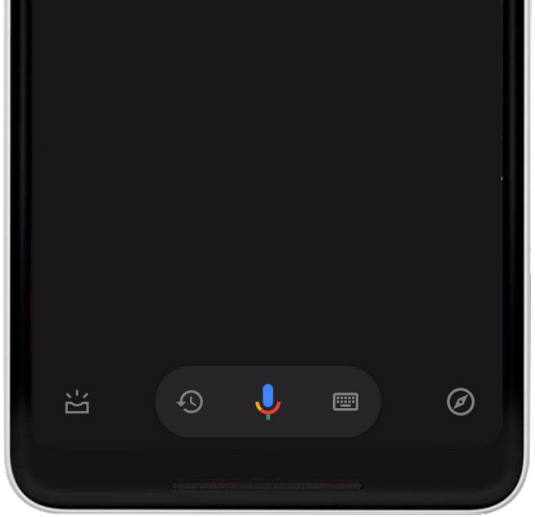





8:00
Dinner with Layla in 30 min
8:30 - 10:00 pm | 50°F

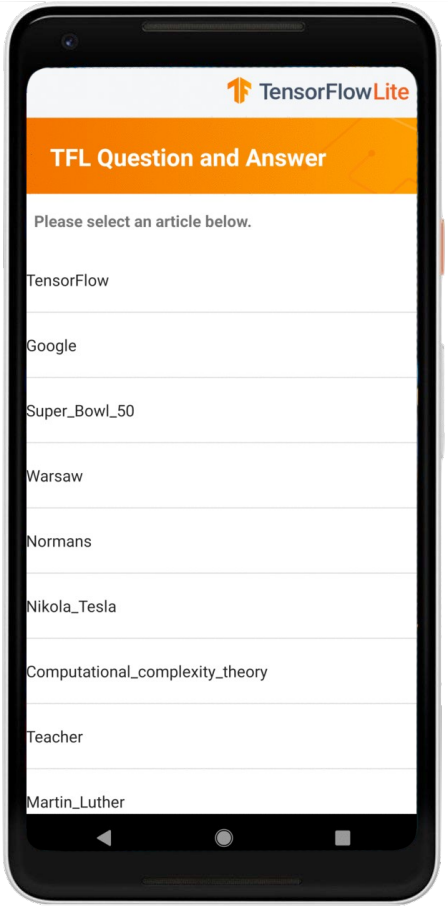

Dinner with Layla in 30 min





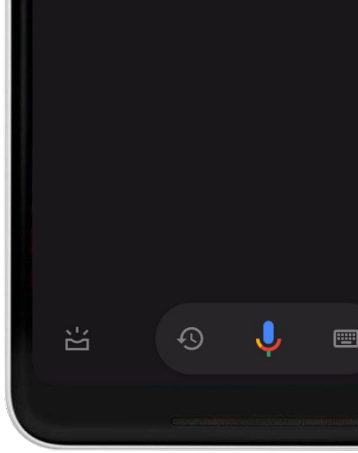

Dinner with Layla in 30 min

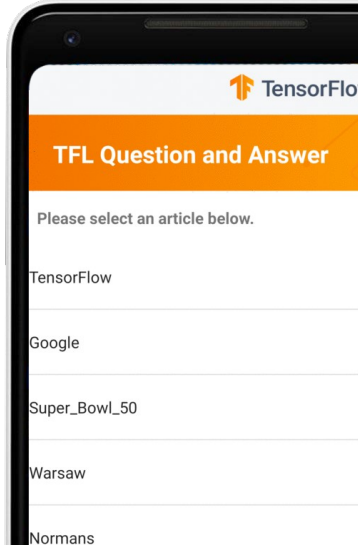



TensorFlowLite
TFL Question and Answer
Please select an article below.
TensorFlow
Google
Super_Bowl_50
Warsaw
Normans
Nikola_Tesla
Computational_complexity_theory
Teacher
Martin_Luther

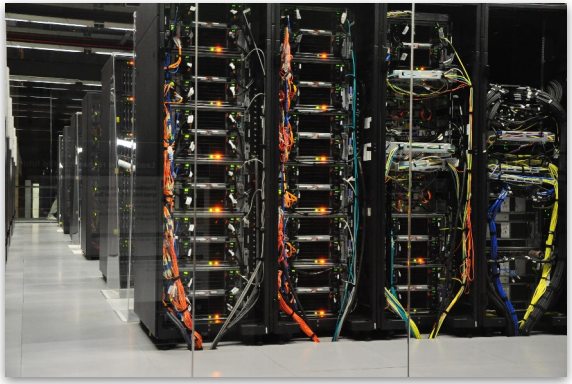







TensorFlo
TFL Question and Answer
Please select an article below.
TensorFlow
Google
Super_Bowl_50
Warsaw
Normans

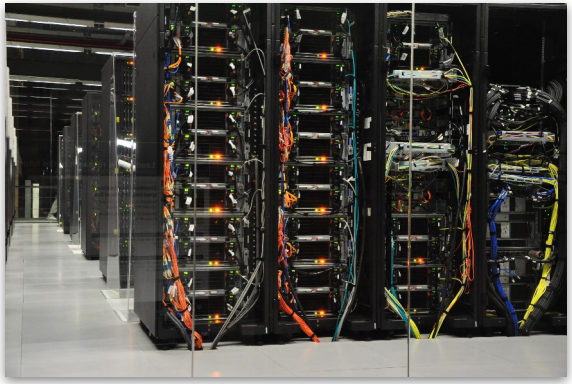
Why?



High power



Why?

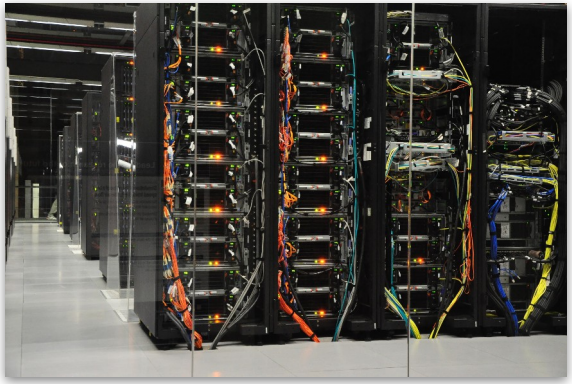


High power



Low power

Why?



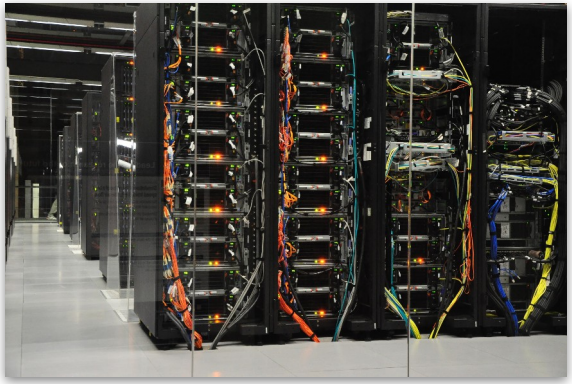
High power

High bandwidth



Low power

Why?



High power

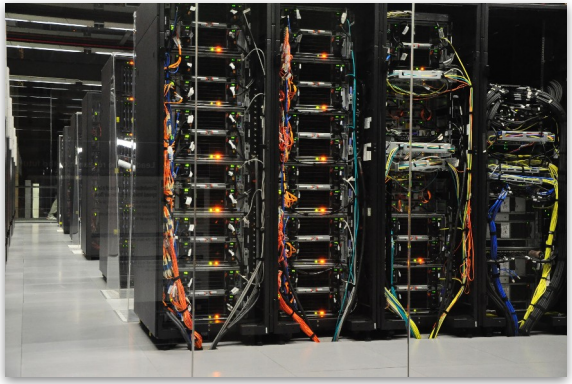
High bandwidth



Low power

Low bandwidth

Why?

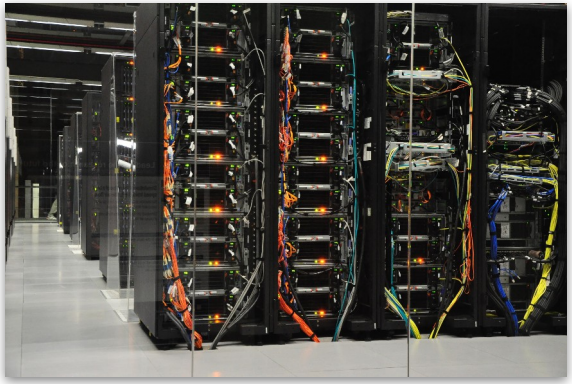


High power
High bandwidth
High latency



Low power
Low bandwidth

Why?

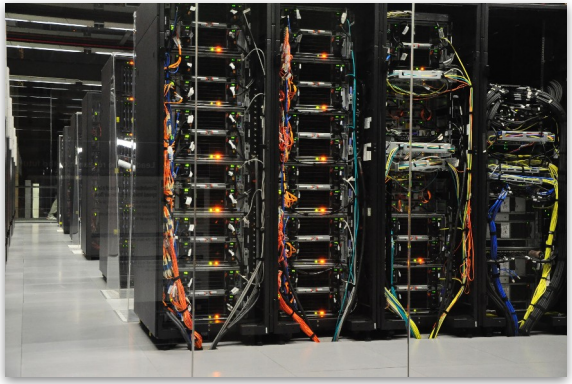


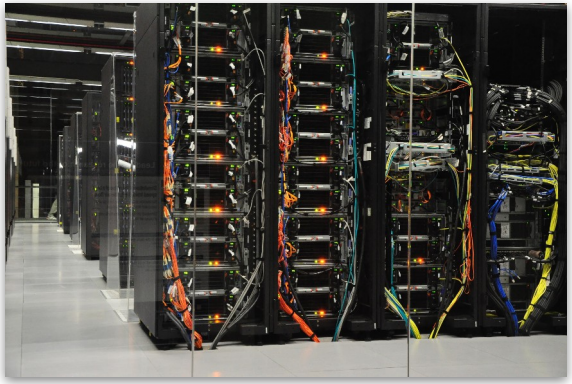
High power
High bandwidth
High latency



Low power
Low bandwidth
Low latency

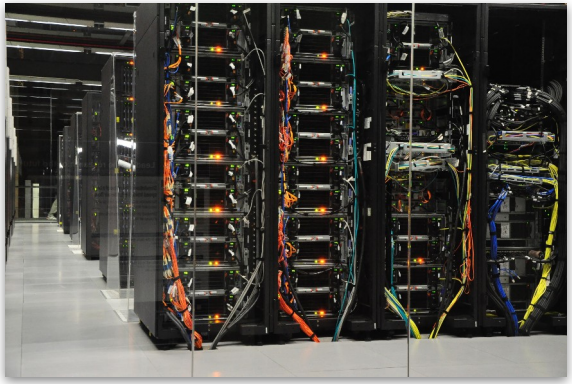

Dinner with Layla in 30 min
ALWAYS
ON?








Dinner with Layla in 30 min











Google Assistant



Endpoint Devices

Google Assistant



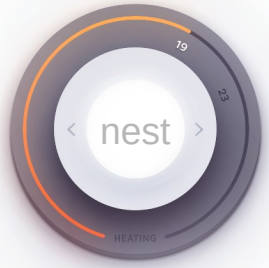
Endpoint Devices



Google Assistant









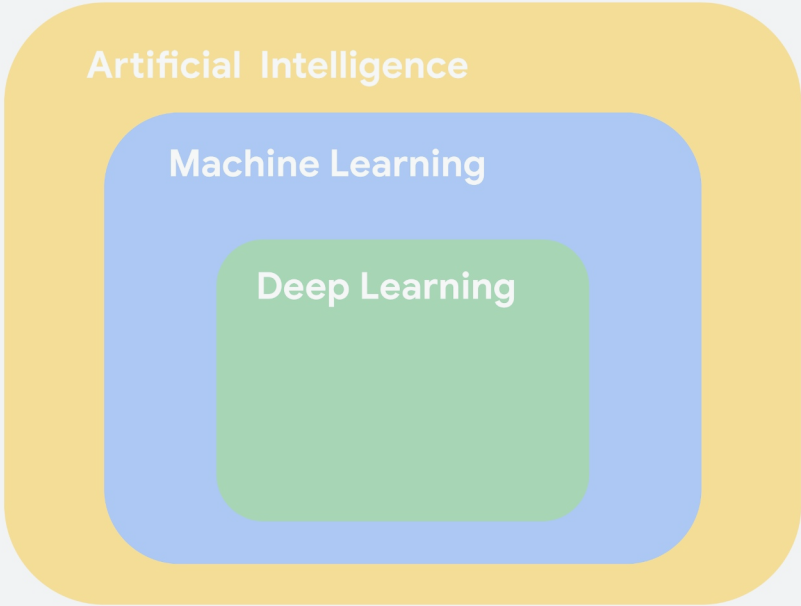




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No Good Data Left Behind

5 Quintillion

bytes of data produced every day by IoT

<1%

of unstructured data is analyzed or used at all

Source: Harvard Business Review, What's Your Data Strategy?, April 18, 2017
Cisco, Internet of Things (IoT) Data Continues to Explode Exponentially. Who Is Using That Data and How?, Feb 5, 2018

Summary

- ML has several diverse applications in the real-world
- ML is increasingly moving from the cloud to endpoint devices
- Endpoint devices are everywhere around us

Half-screen. Show presenter.

Fullscreen. Show presenter.